Mohammad Tabatabaei

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Education

M.Sc. in Complex Systems, Isfahan University of Technology, Iran — Sep 2023 – Jan 2026
GPA: 17.37/20 — *First Class Honors & Full Tuition Scholarship*
Thesis: Impact of Vertex Degree on Exploration-Imitation Dynamics in Complex Systems

B.Sc. in Physics, Shahrekord University, Iran — Sep 2017 – Sep 2022
GPA: 15.57/20 — *Second Class Honors & Full Tuition Scholarship*

Technical Skills

Programming & Data Analysis

- **Python** (Pandas, NetworkX, Scikit-learn, TensorFlow) – data analysis, network science, machine learning, agent-based simulations
- Julia (familiar with basic syntax)
- FORTRAN (familiar with basic syntax)
- R (basic data manipulation and visualization)
- Git, Linux, Bash scripting

Modeling & Simulation Tools

- Agent-based modeling in NetLogo and Python (custom simulations)
- COMSOL Multiphysics
- Quantum Espresso (basic)
- LaTeX (advanced)
- High-Performance Computing (HPC) environments

Research Experience

Impact of Vertex Degree on Exploration-Imitation Dynamics — 2024–2025
M.Sc. Thesis Supervised by Dr. Keivan Aghababaei Samani, Isfahan University of Technology
Extended a social learning model by incorporating degree-based prestige bias into neighbor selection and imitation strength. Simulated four model variants on eight synthetic network topologies.

- Showed that degree-dependent imitation strength amplifies opinion diversity in heterogeneous networks, while degree-biased selection reduces diversity in modular networks.
- Found that combined biases yield intermediate steady-state variances, with effects depending sensitively on topology.
- Established a power-law decay of hub influence in Barabási–Albert networks, demonstrating that denser networks suppress prestige effects.

Entropy-Based Analysis of Centrality Distributions — 2025
Research Project supervised by Prof. Farhad Shahbazi, Isfahan University of Technology
Quantified structural heterogeneity in real and synthetic networks via Shannon entropy of centrality distributions.

- Computed entropy across multiple centralities for real-world networks and matched Erdős–Rényi, configuration, Barabási–Albert, and Watts–Strogatz models.
- Demonstrated that random models exhibit higher, uniform entropy, while real networks show lower, varied entropy; introduced joint entropy of centrality correlations as a higher-order complexity measure.

Modeling Enzyme Network Structure via Centrality Correlations 2025

Research Project supervised by Prof. Farhad Shahbazi, Isfahan University of Technology

Used the correlation matrix of seven centrality measures to compare enzyme interaction networks with classical generative models.

- Analyzed 10 enzyme networks; compared correlation matrices against Barabási–Albert, Watts–Strogatz, and Erdős–Rényi models with matched size and average degree.
- Found Watts–Strogatz model ($p = 0.01$) best reproduced correlation patterns in 9 of 10 cases; proposed hybrid models for improved reconstruction.

Computational Physics in Drug Discovery 2021

B.Sc. Research Project Supervised by Dr. Ali Mokhtari, Shahrekord University

Conducted a comprehensive review of physics-based computational methods in drug discovery, covering both classical and quantum approaches.

- Surveyed structure-based drug design (SBDD) pipelines, protein–ligand docking mechanisms, and force field sensitivity in molecular dynamics simulations.
- Investigated hybrid QM/MM methods for modeling biochemical systems with quantum accuracy in active sites and classical efficiency in bulk environments.
- Synthesized findings into a technical poster presentation; developed proficiency in scientific literature review and interdisciplinary communication.

Research Output

- **Impact of Vertex Degree on Exploration–Imitation Dynamics in Complex Systems**
  Poster – 30th Annual IASBS Meeting on Condensed Matter Physics, Zanjan May 2025
  Manuscript in preparation – Computational study.
- **Mathematical Formulation of Degree-Dependent Exploration-Imitation Dynamics**
  Manuscript in preparation – Analytical extension of the above.
- **Entropy-Based Analysis of Centrality Distributions in Complex Networks**
  Poster – Sharif University of Technology, Tehran May 2026
- **Modeling the Structure of Enzyme Networks through Centrality Correlations**
  Oral Presentation – Iran Physics Conference, Yasouj University Oct 2025

Teaching & Mentoring

Complex Systems (Graduate Workshop) 2025–2026

Workshop Instructor & Content Developer

Led computational and analytical workshop sessions on data analysis and network science. Designed distinct project sets for two consecutive terms (available on GitHub). Graded and provided feedback on all assignments and final projects.

Advanced Statistical Mechanics II (Graduate) 2025

Teaching Assistant

Guided homework and grading; conducted workshops on stochastic simulations and agent-based modeling. Designed and evaluated graded projects (hosted on GitHub).

Critical Phenomena (Graduate) 2026

Teaching Assistant

Assist students with problem-solving and homework; grade assignments.

3rd New Horizons in Physics School (National), Complex Systems Modeling
Module Lead – Isfahan University of Technology Sep 2025
Organized and taught the complex systems module, covering analytical and agent-based approaches to SIR dynamics on Watts–Strogatz networks. Designed project assignments and mentored a student team; received strong positive feedback from program referees.

Workshops & Schools

- **QBronze: Quantum Computing & Programming with Qiskit** Oct 2025
  QWorld – In-person workshop (Diploma QBronze172-16)
- **QPrep: Preparation for Quantum Computing & Programming** Oct 2025
  QWorld – In-person workshop on linear algebra and Python (Diploma QPrep20-15)
- 7-day Online Workshop on **Quantum ESPRESSO** May 2025
  Organized by Scidart Academy and Gdańsk University of Technology
- Winter School on **Physics of Complex Fluids**, IPM, Tehran Feb 2025
- Computational Physics Workshop on **Tensor Networks**, IUT Jan 2024
- 28th Physics School, IASBS, Zanjan Jul 2023
- Second Conference on **Quantum Technologies**, IUT Jun 2023
- Educational Workshop on **Quantum Technologies**, Sharif University Mar 2023

Research Interests

Agent-based modeling on complex networks, stochastic dynamics, and data-driven discovery in physical systems. Emphasis on large-scale simulation and machine learning for fundamental and applied physics, with particular interest in:

- Econophysics and social physics — statistical mechanics of markets, opinion dynamics, and collective behavior
- Mathematical foundations of AI and machine learning — statistical learning theory and physics-inspired approaches
- Biophysics and neuroscience — network approaches to biological complexity, brain connectivity, and neural dynamics

Languages

- **English** – C1 *IELTS Certificate*
- **Persian** – C2 (Native)
- **Arabic** – A1

References

Dr. Keivan Aghababaei Samani – Associate Professor, Complex Systems, IUT (Supervisor)
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Prof. Farhad Shahbazi– Professor, Complex Systems, IUT (Advisor)
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